TUH USED PAPERS OCT/NOV 2

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Code No: 152AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, November/December - 2020
CHEMISTRY
(Common to EEE, CSE, IT)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain various factors that influence the magnitude of crystal field stabilization energy in complex compounds.
b) Explain filling of electrons in F_2 molecule in terms of Molecular orbital theory with neat diagram. [7+8]

2.a) Explain the postulates of Molecular orbital theory.
b) With the help of MO diagram, calculate bond order of O_2 molecule and predict its magnetic behavior. [7+8]

3.a) How municipal water can be treated before distribution for the domestic purpose?
b) Determine the temporary, permanent and total hardness in ppm units for a water sample from Warangal which showed the following analysis:- $Ca(HCO_3)_2$ = 30.4 mg/litre; $Mg(HCO_3)_2$ = 30.2 mg/litre; NaCl = 5.85 mg/litre; $CaCl_2$ = 20.2 mg/litre, $MgSO_4$ = 11.2 mg/litre; Organic matter = 15.5 mg/litre. [7+8]

4.a) Describe the reverse osmosis process for desalination of brackish water with diagram.
b) What is the process of Chlorination? What are the advantages? Explain the significance of breakpoint chlorination. [7+8]

5.a) Explain the following factors that can influence the rate of corrosion:
i) specific volume ratio ii) Position in galvanic series iii) Humidity.
b) Describe the construction and working of calomel electrode. [8+7]

6.a) Calculate the equilibrium constant for the following reaction at 298K.
$Cu + Cl_2 \text{ ------} \rightarrow CuCl_2$ $E^0_{Cu+2/Cu}$= 0.34 V : $E^0_{Cl2/2Cl-}$ = 1.36 V.
b) What is the principle involved in the cathodic protection? Discuss the sacrificial anodic and impressed current methods. [7+8]

7.a) Explain Markownikoff's addition products in addition of HBr to propene.
b) What are conformers? Discuss the conformational analysis with reference to n-Butane. [7+8]

8.a) Discuss various factors that influence chemical shift.
b) Give an account of various modes of fundamental vibrations. [7+8]

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